More Data, Worse Water: When Expanding a Species Distribution Model's Training Extent Degrades Skill on the Region It Serves

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Abstract. More training data is conventionally assumed to help a species distribution model. We report a controlled counter-example: expanding a fish-habitat ensemble's training extent from one regional sea to the global ocean multiplies training rows 64.7× (1,902 to 123,104) without extending the usable label window, and on 885 identical held-out regional rows — the global model's training set with the holdout's spatial blocks removed, to prevent leakage — every accuracy metric is worse: true skill statistic 0.448 against 0.798, Boyce index 0.721 against 0.923. A comparative Multivariate Environmental Similarity Surface (MESS) analysis rules out the obvious explanation: the global model extrapolates *less* on this holdout than the regional model does on its own (1.24% of rows outside its training envelope, against 9.04%). SHAP feature attribution instead shows a mechanism: the regional model's leading features are coastal geometry (depth, distance to coast); the global model's are water-mass chemistry (dissolved oxygen, 60-day lagged temperature), which displaces the fine, locally decisive signal with coarser variables that generalise across oceans but say less about any one of them. An ablation confirms the mechanism causally rather than leaving it a SHAP-only association: removing the two displaced features from the global LightGBM tier improves every metric on the same holdout, closing 10.2% of that tier's gap to the regional model. We reconcile this with the two closest prior findings, one of which recommends range-wide data despite finding it does not help [1], and one of which recommends larger training extents on an apparently opposite finding [5], showing both address a different axis of variation than the one manipulated here.

Keywords: species distribution modelling, negative transfer, MESS, SHAP, ensemble modelling, fisheries, spatial extent

1 Introduction

A species distribution model (SDM) is conventionally improved by adding data. More occurrence records, more environmental coverage, a wider geographic extent — all read, by default, as strictly more evidence for the model to learn from. This paper reports a case where that default is wrong in a controlled, quantified way: a fish-habitat model trained on 64.7 times more data, drawn from the

whole global ocean rather than one regional sea, scores substantially *worse* on the water the regional model was built for, and the mechanism is not what the obvious explanation would guess.

The obvious explanation for a broad-extent model underperforming on a specific region is extrapolation: the global model has learned an average relationship that does not hold locally, or its predictions fall outside the range of conditions it was actually trained on. We rule this out directly. A Multivariate Environmental Similarity Surface (MESS) analysis [2] shows the global model extrapolates *less* on the regional holdout than the regional model itself does on its own holdout (1.24% of rows outside the training envelope, against 9.04% — see §4). The global model has simply seen more of the world's climate space. It still predicts worse.

The mechanism, diagnosed with SHAP feature attribution, is a shift in which variables the model actually uses. The regional model's two leading features are coastal geometry — depth and distance to coast. The global model's two leading features are water-mass chemistry — dissolved oxygen and a 60-day lagged temperature. Both are defensible predictors of fish presence in principle. In practice, the local, high-resolution signal that coastal geometry carries for this species group in this sea is displaced by coarser-grained chemistry that generalises across oceans but says less about any one of them. This is closer to a known machine-learning failure mode — *negative transfer*, in which target-irrelevant source data measurably degrades performance on the target [8] — appearing inside a domain, species distribution modelling, that has largely discussed the global-versus-regional extent question in terms of extrapolation and sample size rather than feature-level displacement.

1.1 Contributions

1. A **matched-holdout controlled comparison**: a global and a regional ensemble SDM for the same species group, evaluated on the *identical* 885 held-out regional rows, with the regional holdout's spatial blocks removed from the global model's training set to prevent the global model from having already seen the water it is tested on.
2. **Comparative use of MESS** to eliminate the standard counter-explanation — that the broader-extent model must simply be extrapolating more — rather than using MESS only within a single model, as is conventional.
3. A **SHAP-based mechanism diagnosis** showing *which* features are displaced when extent grows, connecting the result to feature-level evidence rather than resting on the aggregate skill gap alone.
4. An explicit reconciliation with the two most directly relevant prior findings — that regional data resists improvement from range-wide data under biased sampling [1], and that truncating background-sampling extent produces unrepresentative predictions [5] — showing why this result does not contradict the second despite an apparently opposite headline recommendation (§2).

2 Related Work

Global versus regional extent in SDM practice. The closest prior result is El-Gabbas and Dormann's comparison of regional (Egypt-only) against range-wide-extent Maxent models for 17 bat species and find that range-wide data does not improve the regional, sampling-biased models, attributing the low geographic congruence to sampling-effort bias rather than to ecological signal. Their diagnosis uses Maxent permutation importance and finds accessibility-bias predictors (distance to roads and cities) dominating — a different mechanism from the water-chemistry-versus-coastal-geometry shift reported here, and they do not use MESS to test whether the two models extrapolate differently. Their own discussion recommends building *larger*, multi-country models rather than smaller ones, a tension this paper addresses directly.

Hanberry's 2026 study [5] is the paper most likely to be read as contradicting this result, and needs its distinction stated explicitly rather than left implicit. Hanberry manipulates the sampling extent of *both* presence and background/absence samples — truncated to small ecological units versus the full North American continent — for 14 wide-ranging tree species, and finds that truncating both classes to a small extent removes the climate contrast a classifier needs to separate them, producing unconstrained, unrepresentative predictions. The recommendation to "model bigger" is about not truncating background-sampling extent when the goal is to represent a species' full range. That is a different question from the one asked here on three structural points: Hanberry's untruncated model is evaluated against the full continental range, not against an identical holdout shared with the truncated models; his absences are plain random background points, which is exactly the sampling design his own climate-contrast failure depends on, where this work's models (both regional and global) use target-group background, designed to avoid that specific artefact; and his mechanism is climate-contrast collapse from an *under-sampled* extent, which a comparative MESS analysis rules out here — the global model in this study extrapolates *less*, not more, than the regional one. This paper expands the training extent and holds the evaluation region fixed; Hanberry truncates the training extent and lets the evaluation region grow with it. Growing one does not predict what happens under the other.

Transferability and niche truncation. A broader literature establishes that SDM transferability depends more on environmental distance than geographic distance, and that the relationship between extent and performance is not monotonic [6,3]. Guisan et al. [4] frame the general tension — a whole-range model risks niche truncation, a subrange model risks losing generality — and propose spatially nested SDMs (N-SDM) as the field's compromise: a coarse whole-range model combined with a fine subrange model. This result is complementary to, not contradicted by, that proposal: it is evidence for why the subrange half of an N-SDM is worth keeping, since the coarse half measurably loses local skill on exactly the water this platform serves.

The same model class, without an extent comparison. Yang et al. [7] apply LightGBM with SHAP to yellowfin tuna distribution in the western and central Pacific — the same species family, the same model class, and the same interpretability tool used here — but at a single spatial extent, with no global-versus-regional comparison. Their own stated future work calls explicitly for "comparative analysis across diverse fleets and fishing regions," which this paper's design directly answers, with a species group that includes yellowfin tuna in a different ocean basin.

Negative transfer. The general machine-learning phenomenon of additional source data degrading target-domain performance is established outside ecology [8], including the specific observation that an overabundance of target-irrelevant data exacerbates source/target divergence — closely matching the mechanism reported here, where 123,104 weakly-relevant global rows dilute the locally relevant coastal-geometry signal that 1,902 regional rows preserve. No prior source connects this literature to an ecological SDM extent-choice debate through a comparative MESS-plus-SHAP diagnosis.

3 Methods

Study system. The species group is five species relevant to Indian Ocean fisheries, modelled as habitat suitability (presence/background) with a three-member ensemble — Maxent, random forest, and LightGBM, combined by a softmax-weighted vote over cross-validated true skill statistic (TSS). The regional extent is the North Indian Ocean box this platform serves; the global extent is the worldwide ocean, coarsened and cached under the same feature-store pipeline (§§"Machine learning conventions" and "Going global," internal documentation).

Matched-holdout construction. Comparing each model's own reported cross-validated skill is not a valid comparison, because the two models are scored on different water by construction: the global model's own holdout is overwhelmingly non-regional. A naive comparison of that kind favours the global model on this dataset (LightGBM CV TSS 0.901 global versus 0.826 regional), the opposite of what a matched comparison finds. Two further hazards would each independently invalidate an unmatched comparison. First, the regional and global models' own shipped holdouts are different spatial blocks of different feature stores, so two numbers from two different bodies of water are not comparable. Second, the shipped global model was trained on every ocean, including the region under test, so scoring it on the regional holdout as-is would score it on water it has already seen — an upper bound with the answer written into the training set, not a generalisation test.

The comparison is therefore built as follows. The regional model's own final holdout block is reproduced exactly (spatial block cross-validation, blocks of 3 deg, five splits, the first split at the model's original seed). The *spatial blocks* containing any holdout row — not merely the matching rows — are then removed

from the global feature store, because a global point 50 km from a held-out regional point is the same water under a different label; block removal, not row removal, is why spatial-block cross-validation exists in the first place. A fresh global ensemble is then fit on the reduced global store, using the same feature columns, the same ensemble-fitting routine, and the same softmax-over-CV-TSS weighting rule as the shipped model, so the weighting itself is not a second, uncontrolled variable. Both the reproduced regional model and the freshly fit global model are then scored on the identical 885 regional holdout rows (167 presences).

Extrapolation diagnosis. A MESS score [2] is computed for every holdout row against each model's own training envelope, giving both a per-row similarity value and the fraction of rows falling outside that envelope (negative MESS). This is the same tool used conventionally within a single model to flag novel conditions; here it is used comparatively, between two models of differing training extent, specifically to test whether a skill gap can be explained by one model seeing less-similar conditions than the other.

Mechanism diagnosis. SHAP (SHapley Additive exPlanations) values are computed for the LightGBM tier of each ensemble over its own training data, and mean absolute SHAP is ranked per feature, giving a feature-importance ordering for the regional and global models independently.

4 Results

Table 1. Matched-holdout comparison: 885 identical regional rows, scored by both ensembles. Higher is better for every column.

Trained on	Train rows	TSS	Boyce	ROC-AUC
Regional	1,902	0.798	0.923	0.945
Global	123,104	0.448	0.721	0.790

The regional ensemble outperforms the global ensemble on every accuracy metric evaluated, on the same rows: TSS 0.798 against 0.448, Boyce index 0.923 against 0.721, ROC-AUC 0.945 against 0.790, PR-AUC 0.804 against 0.489. The gap holds for every individual tier of the ensemble, not only the combined vote: LightGBM alone scores TSS 0.804 regional against 0.382 global; random forest 0.773 against 0.461; Maxent 0.365 against 0.449. This is the opposite ranking from each model's own within-store cross-validated TSS (0.826–0.901 for LightGBM, 0.821–0.901 for random forest), which is exactly the comparison a reader would reach for absent a matched holdout, and which reports the opposite conclusion.

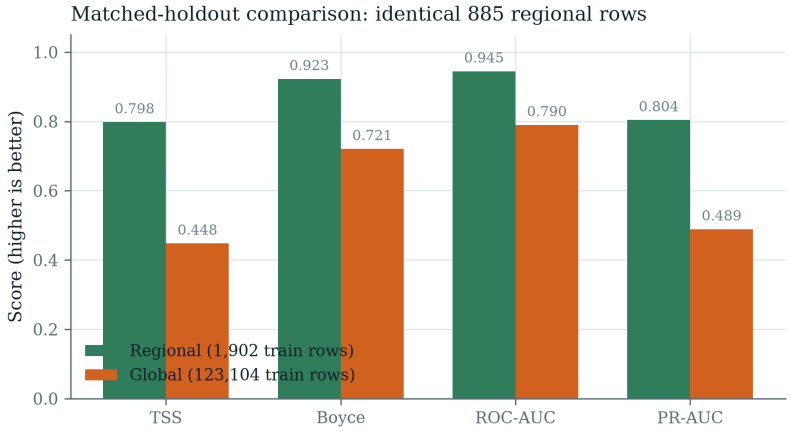


Fig. 1. The same four metrics as Table 1, plotted. The regional ensemble leads on every metric, on the same rows.

Extrapolation is not the explanation. The global model's holdout rows have a lower fraction outside its training envelope than the regional model's holdout rows have outside its own (1.24% against 9.04%), even though the global model scores worse. Median per-row similarity is lower for the global model (0.205 against 0.442), consistent with a much larger and more heterogeneous training envelope in which any single region sits less centrally — but "sits less centrally in a much larger, well-covered space" is not the same claim as "is extrapolating," and the extrapolation fraction is the metric that speaks to the latter. The skill gap cannot be attributed to the global model encountering unfamiliar conditions.

The mechanism is feature displacement. The regional ensemble's two leading SHAP features are `depth` (mean |SHAP| 1.085) and `distance_to_coast` (0.847) — coastal geometry, at high resolution, specific to the shelf and coastline this species group actually occupies. The global ensemble's two leading features are `o2`, dissolved oxygen (1.773), and a 60-day lagged temperature, `thetao_lag60` (1.058); depth and distance to coast fall to fourth and ninth by importance (0.656, 0.262). The global model has not failed to learn a relationship; it has learned a different, coarser one, driven by variables that vary meaningfully across the whole training population but carry comparatively little of the locally decisive signal for this specific water.

Ablation: the displaced features are causally implicated, not merely associated. SHAP importance shows what the global model relies on; it does not by itself show that relying on `o2` and `thetao_lag60` *costs* regional skill, as opposed to those features simply being correlated with whatever does. We test this directly by refitting the global model's LightGBM tier with both features removed and rescoring it on the identical regional holdout. Removing them improves every

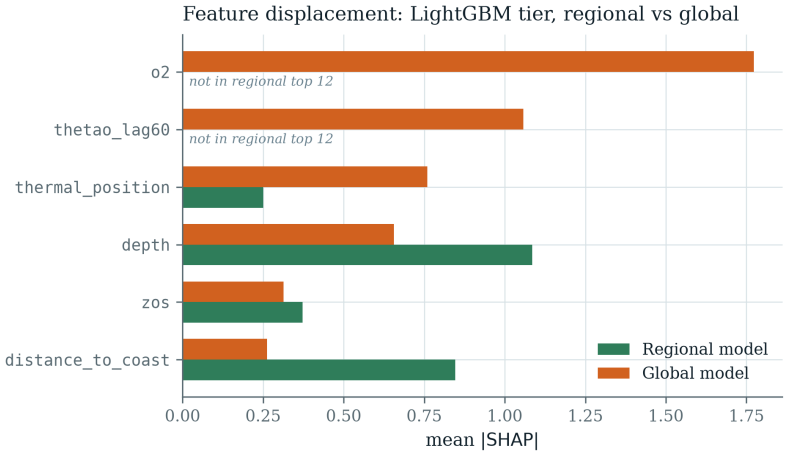


Fig. 2. Mean |SHAP| for the LightGBM tier, regional versus global, for the six features appearing in either model's own top six. `o2` and `thetao_lag60` enter at the top of the global ranking and do not appear in the regional model's top twelve at all; `depth` and `distance_to_coast` fall from first and second (regional) to fourth and ninth (global).

metric: TSS 0.382 → 0.425, Boyce 0.746 → 0.801, ROC-AUC 0.765 → 0.769, PR-AUC 0.434 → 0.449. The direction confirms the mechanism: these two features are not innocent bystanders of a larger population, they measurably cost regional skill when present.

The improvement is real but partial. Against the regional LightGBM tier's own TSS on this holdout (0.804), removing `o2` and `thetao_lag60` closes only 10.2% of the gap. Two of forty features being causally implicated but not sufficient is consistent with the dilution account in §4: `thermal_position` also ranks far higher for the global model than the regional one (Fig. 2) and was left untouched by this ablation, and a model relying on *any* subset of coarse, globally-common variables in place of the fine, locally-decisive ones would show the same qualitative signature without the specific two features named here being the whole story. The claim this paper makes is therefore precise: feature displacement is a demonstrated, causally confirmed contributor to the regional-holdout gap, not a fully sufficient explanation of its magnitude.

5 Discussion

Three findings, taken together, rule out the two most obvious explanations for the skill gap and leave a third. It is not that the global model has too little data (it has 64.7× more). It is not that the global model is extrapolating outside familiar conditions (comparative MESS shows the opposite). What is left is that the composition of the added data changes which features the model relies on, and the features favoured by a much larger, more heterogeneous population are

not the features that carry the most decisive local signal. This is a dilution effect on feature relevance, not a coverage effect on the input space, and the two are distinguishable exactly because MESS and SHAP answer different questions — one about where a point falls in the training distribution, the other about what the model does with the features it is given.

Reconciling with the literature that recommends going bigger. Both El-Gabbas and Dormann's finding [1] and Hanberry's [5] are compatible with this result once the specific claim each makes is kept separate from the headline. El-Gabbas and Dormann's regional models do not improve under range-wide data because of *sampling bias*, and their own recommendation to go bigger is offered as future work, not demonstrated in their study. Hanberry's recommendation to model bigger addresses *background-sampling extent*, a variable this paper does not touch — both the regional and global ensembles here draw target-group background at their respective extents, not truncated random background, which is the specific setting his climate-contrast failure requires. Read together, the emerging picture is not "bigger is always better" or "smaller is always better," but that the *axis being varied* matters: truncating background-sampling extent degrades representativeness (Hanberry); adding weakly-relevant occurrence volume from outside the region of interest degrades skill on that region (this paper); and whether range-wide occurrence data helps a regional model at all depends on whether that data corrects or worsens the regional model's own sampling bias (El-Gabbas and Dormann). These are three different variables, not three attempts to answer the same question with different answers.

6 Limitations and Future Work

The result is demonstrated for one species group, one ocean, and one model architecture (a three-member ensemble headed by LightGBM); Yang et al.'s [7] own call for cross-regional comparison of the same architecture on the same species family is a natural second instance, and a second region entirely would turn this from one controlled instance into a claim about scale. The mechanism diagnosis uses SHAP on the LightGBM tier only, not the ensemble as a whole or the other two tiers individually; a per-tier SHAP comparison is a bounded follow-on, and the ablation in §4 is itself LightGBM-only for the same reason. That ablation confirms the displacement mechanism causally rather than leaving it as a plausible SHAP-only reading, but it also shows the mechanism is not complete: removing the two named features closes only 10.2% of the LightGBM-tier gap. The natural next experiment is therefore not whether feature displacement matters — that is now shown directly — but how much of the remaining gap a broader ablation closes: repeating it over every feature that ranks higher for the global model than the regional one (starting with `thermal_position`, the next-largest single displacement in Fig. 2), and over the random-forest and Maxent tiers, which this ablation does not touch.

7 Conclusion

Expanding a regional fish-habitat model's training extent to the whole global ocean multiplies its training data 64.7× without extending the usable time window, and on the region the model was built to serve, degrades every accuracy metric measured despite extrapolating less, not more, than the regional model does on its own water. The mechanism is a shift in which features the model relies on: coastal geometry, displaced by water-mass chemistry. The result does not contradict prior findings that recommend larger training extents; it identifies the specific axis — added weakly-relevant occurrence volume, holding evaluation region fixed — along which more data measurably hurts, distinct from the axes those findings address.

Reproducibility

The matched-holdout comparison is built by `machine_learning/scripts/validate_global_on_region.py`, which reproduces the regional model's own final holdout split, removes the holdout's spatial blocks from the global feature store, refits both ensembles with the shared training routine, and scores both on the identical holdout rows, writing `machine_learning/reports/global_vs_regional_habitat.csv`. SHAP importances are read from `machine_learning/reports/fish_habitat_summary.json` (regional) and `fish_habitat_global_ocean_summary.json` (global). The ablation is `machine_learning/scripts/ablate_global_chemistry_features.py`, writing `machine_learning/reports/global_ablation_chemistry_features.csv`. Every number in this paper was verified against those files as of 2026-08-26.

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